

Chapter 68 — Integrated Men's Health: Androgen Deficiency, Cardiovascular Risk, and Metabolic Syndrome

Campbell-Walsh-Wein Urology · Part VI Reproductive and Sexual Function · Thomas Walsh and Joseph P. Alukal

- The **multidisciplinary men's health centre** is designed to **maximise care of male patients** while **increasing convenience** of obtaining it; such centres are increasingly prevalent.
- Dual role: it raises the **quality of preventive and other care**, and it acts as a **front door for new male patients** into urology practice; proper design drives satisfaction, retention and quality.
- **Urologists are best placed to create and run these centres**, because the complaints that bring men into the health system as first-time users are **often urologic**.
- The multidisciplinary component (primary care, cardiology, gastroenterology) requires the urologist to **learn specialties beyond urology**.

ISSUES OF MALE UROLOGIC DISEASE ACROSS ALL AGES

Introduction and General Considerations

- Prevalence of urologic and nonurologic disease in men **by age is well described**.
- The **Urologic Diseases in America project** quantified the incidence of benign and malignant urologic disease and the burden on urology providers (annual visits, procedural volume).
- Combined with primary care and cardiology visit data, **visit volumes for middle-aged and older men are reasonably predictable**.
- Young adults: data less robust, but **male infertility, erectile dysfunction (ED) and other sexual complaints, and testis cancer** generate visits at steady rates.
- Nonurologic malignancy is similarly predictable; **endocrinopathy, cardiac prevention or early cardiac disease and other preventive measures are variable**.
- Accurate prediction of need governs **provider and resource planning**.

ADOLESCENCE/YOUNG ADULTHOOD (18–39 YEARS)

- **Urologic complaints**, managed by the urologist or a urology-trained advanced practice provider (APP): ED, rule out hypogonadism, testis and pelvic pain, sexually transmitted infection exposure or urethral pain, Peyronie disease, premature ejaculation, fertility, testis mass and testis cancer, varicocele.
- **On-site testing**: urinalysis, serum hormones and other blood tests, Doppler ultrasound. **"Rule out hypogonadism" is a commonly occurring visit type**.
- **Nonurologic disease**: endocrinopathy (diabetes, thyroid), hypertension, hyperlipidaemia, psychiatry complaints, obesity, haematologic (leukaemia, lymphoma) and solid (colon, sarcoma) malignancy, cardiac prevention, upper respiratory tract infection, well visits, cosmetic and dermatology care and vaccination — seen by a family or internal medicine physician, nurse practitioner or physician assistant, some by telemedicine.
- **Prevention**: smoking cessation, moderation of alcohol and overeating, counselling on exercise and sleep.

MIDDLE AGE (40–64 YEARS)

- **Prostate cancer screening (Chapters 153 and 157) and colon cancer screening begin**; easy gastroenterology referral matters, and an in-house gastroenterologist may be justified by volume.

- **Cardiac screening** and management of active coronary artery disease (CAD) gain importance, as does **pulmonology**, because chronic obstructive pulmonary disease (COPD) and lung cancer incidence rise (National Lung Screening Trial).
- Endocrinology and bariatric referral are provided according to volume and regional demographics.

OLD AGE (AGES 65+)

- Priorities: **fall prevention**, neurologic (including degenerative) disease, **hearing and vision loss**, nutrition and social issues. Most centres cannot staff each in house, so they rely on **established referral patterns**.

MEN'S HEALTH CENTRE DESIGN BY AGE BAND

	18–39 years	40–64 years	65+ years
Dominant need	Sexual, reproductive, testicular complaints; prevention	Cancer screening begins ; active disease management	Geriatric syndromes; coordination
Primary care posture	Almost wholly preventive	Shifts to active management; gatekeeper to specialists	Changing preventive focus
Key partners	Sports medicine, physiatry/physical therapy, orthopaedics; mental health (all ages)	Gastroenterology, cardiology, endocrinology, bariatrics, pulmonology	Geriatrics , neurology, audiology, ophthalmology, nutrition, social care
Delivery model	Urologist or APP; telemedicine	In-house specialist if volume justifies	Referral network; nurse navigator in high-volume centres

Concept — Centre design follows **predictable age-specific demand**: the focus moves from prevention to disease management to coordination, and in-house versus referral provision is decided by volume.

UTILISATION OF ADVANCED PRACTICE PROVIDERS AND TELEMEDICINE

- APP care is an appropriate **adjunct to both urologic and nonurologic care** at all ages; **essential** in centres serving a large geographic or small population region.
- A **specifically trained APP can deliver both** — adjunct to the urologist and independent primary care.
- **Telemedicine** is vital in geographically large, population-sparse regions; **billing and reimbursement changes after the SARS-CoV-2 pandemic** make it financially justifiable in almost all practices.

APPROACH TO PRIMARY CARE OF THE MALE PATIENT

- Core tasks: **prevention, early diagnosis, management of chronic conditions** — methods have changed, goals have not; achieving them improves longevity, quality of life and wellness.
- **Wellness as the mere absence of disease is a passé concept.**
- **Personalised medicine** rests on the patient's genetics and phenotype and, equally, on the **patient's own health goals**.
- **Shared decision making** extends beyond prostate cancer screening and management; the "one size fits all" approach is gone.

THE LONGEVITY GAP BETWEEN MALES AND FEMALES

- Life expectancy is the **yardstick of medical progress** and partly determines national health-system rankings; the **United States ranks poorly** among developed nations, and the reasons are debated.
- **Females outlive males**: the gap has narrowed but remains significant, and males have higher mortality across most leading causes of death (detailed below).

- The causes are poorly understood. **Hormonal hypotheses** — adverse physiologic effects of testosterone, protective (notably cardiovascular) effects of oestrogen — address only one explanation; **behavioural, environmental and risk-taking differences** may contribute substantially. In sum, being male confers a **high risk of early mortality**.

MALE VERSUS FEMALE: WHERE THE MORTALITY GAP LIES

Domain	Male disadvantage
Life expectancy (National Vital Statistics System, 2016)	At birth males 76.1 vs females 81.1 years (78.6 overall); at 65, 18.0 vs 20.6 further years
Leading causes of death	Higher in 12 of the 15 , 5 of them more than double; unchanged to 2020 (pre-pandemic)
Suicide	Largest gap — more than threefold
Coronary disease	Higher prevalence; myocardial infarction (MI) more often diagnosed at all ages before 75
Type 2 diabetes mellitus (T2DM)	Rising since the early 1990s and faster in males — gap widening
Cancer	Lung cancer excess partly explains the all-cancer gap; colon cancer earlier, commoner, more advanced at diagnosis; higher mortality from pancreatic cancer, leukaemia, non-Hodgkin lymphoma, liver and intrahepatic bile duct cancer
Prostate cancer	Only US malignancy with rising stage IV at diagnosis over the past 5 years (American Cancer Society) → a screening discussion with every men's health patient is vital
COVID-19	Consistently higher mortality; effect on updated statistics not yet known

Concept — *The longevity gap is a **male excess across most causes of death**, not a single disease, and its causes are largely unexplained — which is precisely why men's health must be multidisciplinary.*

SCREENING FOR UROLOGIC DISEASE

- Validated questionnaires (Chapters 72 and 150) are core tools for the urologist evaluating men.
- A patient presenting with a **sexual health complaint often has a meaningful voiding complaint, and vice versa** — screen both.
- **Prostate, bladder and upper tract tumours**: screened with prostate-specific antigen (PSA) and urinalysis per guidelines; the urologist must **deploy and correctly interpret** these tests.

VALIDATED SCREENING QUESTIONNAIRES

Instrument	Domain screened
Sexual Health Inventory for Men (SHIM)	Erectile function
Androgen Deficiency in the Aging Male (ADAM)	Symptoms of androgen deficiency
International Prostate Symptom Score (IPSS)	Lower urinary tract symptoms

CARDIOVASCULAR PREVENTION AND SCREENING

Burden and global risk assessment

- Apart from COVID-19 in recent years, **heart disease remains the leading cause of death in the United States**.
- **Family history and genetic predisposition** are risk factors but do not account for all of a person's risk.
- **American College of Cardiology/American Heart Association (ACC/AHA) Atherosclerotic Cardiovascular Disease (ASCVD) Risk Calculator**: inputs blood pressure, lipids, smoking and diabetes status, statin and

aspirin therapy → cardiovascular event risk; issues ACC/AHA statin recommendations and low-density lipoprotein (LDL) and high-density lipoprotein (HDL) targets.

- Limitations: **no family-history variable**; cannot capture **diabetes type, duration or glycaemic control**.
- Diabetes control is central to preventive cardiology but usually **outside the centre's scope** unless the primary care provider is facile → close **endocrinology** relationship.

Lipids and statins

- **Primary cardiovascular disease (CVD) prevention** = identifying and modifying risk factors before overt disease; the task of the primary care physician or APP.
- **LDL cholesterol (LDL-C) correlates consistently with cardiovascular events**, and modifying it changes future risk.
- **Only LDL-C lowering benefits patients without overt CVD**; raising HDL-C or correcting triglycerides confers no benefit in the absence of heart disease.
- **Statins (HMG-CoA reductase inhibitors)** are the best-established class; early trials of pravastatin, lovastatin and rosuvastatin reduced **first cardiovascular event and even all-cause mortality in both sexes**; meta-analysis shows strong reductions in events and mortality **with and without established CVD**.
- **Cardiovascular risk discussion and lipid screening from as early as age 20**.
- Elevated LDL-C in primary prevention: **moderate-intensity statin plus diet and lifestyle**. Rationale: **no randomised trial compares moderate with high intensity** in primary prevention; the extra LDL-C reduction from high intensity is small; moderate intensity is better tolerated.

STATIN DECISIONS IN PRIMARY PREVENTION (ACC/AHA)

Situation	Recommendation
LDL-C ≥ 190 mg/dL	High-intensity statin ; consider added nonstatin therapy
LDL-C 70–189, 10-year ASCVD risk $\geq 7.5\%$	Moderate- to high-intensity statin
LDL-C 70–189, risk 5–7.5%	Moderate-intensity statin may be started
Statin-intolerant, LDL-C < 190 , no CVD	Conservative: diet and lifestyle
Statin-intolerant, 10-year risk $> 20\%$	Consider ezetimibe or proprotein convertase subtilisin/kexin type 9 (PCSK9) inhibitor ; long-term primary-prevention safety and efficacy not established by randomised trials
Target	LDL-C as low as 50 mg/dL

STATIN INITIATION THRESHOLD: GUIDELINE BODIES COMPARED

	ACC/AHA	U.S. Preventive Services Task Force (USPSTF); National Institute for Health and Care Excellence (NICE, UK); European Society of Cardiology/European Atherosclerosis Society (ESC/EAS) 2016
10-year ASCVD risk to start statin	$\geq 7.5\%$ (5–7.5% moderate intensity)	$\geq 10\%$
Posture	More aggressive	More conservative
Common ground	LDL-C control is effective primary prevention	

Coronary artery calcification scoring

- **Coronary artery calcification (CAC) scoring** measures atherosclerotic burden noninvasively by **multidetector-row computed tomography**, best used in **asymptomatic patients over 40** to exclude subclinical disease.

- **Highly sensitive** for coronary stenosis $\geq 50\%$ but **only moderately specific**, especially over 60, when calcification occurs without CAD; higher scores are more specific.
- **A zero score carries a very low (<1%) likelihood of coronary atherosclerosis**, so CAC is strongest as a negative predictor.

WHERE CAC SCORING CHANGES MANAGEMENT

Group	Value of CAC
Low risk	Little — unlikely to be positive
Intermediate risk	Greatest — risk stratification
High risk	Treated aggressively anyway; confirms atherosclerosis without invasive coronary or CT coronary angiography and lowers the threshold for aggressive management
Resistant to drug therapy	A positive score persuades the patient of the need for treatment
Diabetic	Zero score may not predict long-term mortality

Genetic and novel markers

TWO APPROACHES TO GENETIC TESTING

Genomic screening	Expression profiling
Scans the genome for genes predicting future disease	Measures genes over- or under-expressed relative to expectation

- **Corus CAD Gene Expression Test** (expression profiling): blood test for expression of CVD-associated genes; aims to detect atherosclerotic heart disease **without invasive coronary angiography**, which is often negative in patients meeting criteria for it. Not extensively studied in diverse populations; early performance good.
- Indication: **atypical chest pain with low suspicion of CAD — a reassurance test, not a positive marker.**
- **Lipoprotein(a) [Lp(a)]**: near-linear relationship with cardiovascular event risk; **unaffected by statins.** Lowering agents (nicotinic acid, PCSK9 antibodies) have **no proven clinically significant effect on CVD risk**; possible future modifiable factor.

Hypertension

- **Hypertension is the most important modifiable risk factor for heart disease.** ACC/AHA definitions updated in **2017**, broadening the definition and lowering treatment thresholds.

ACC/AHA 2017 BLOOD PRESSURE CATEGORIES

Category	Systolic (mm Hg)		Diastolic (mm Hg)
Normal	<120	and	<80
Elevated	120–129	and	<80
Stage 1	130–139	or	80–89
Stage 2	≥ 140	or	≥ 90

- **Risk factors for primary (essential) hypertension:** advanced age, obesity, family history, race, kidney disease, high-sodium diet, alcohol, physical inactivity.
- **Screening:** **annually** in all adults 18 and over without prior hypertension; **at least six-monthly** with risk factors or observed elevated pressure; more often at highest risk.
- **Elevated blood pressure or stage 1** → nonpharmacologic: dietary salt restriction, weight loss, exercise, potassium supplementation, **Dietary Approaches to Stop Hypertension (DASH) diet**, limited alcohol.

- **PREMIER trial:** comprehensive lifestyle modification reduced the rate of hypertension and antihypertensive drug use, though not significantly.
- **Drug therapy: stage 2, or stage 1 plus age ≥ 65 , T2DM, established CVD, chronic kidney disease (CKD), or 10-year ASCVD risk $\geq 10\%$.**
- Treatment reduces relative risk of **heart failure, stroke and MI** versus placebo, and prevents left ventricular hypertrophy, systolic and diastolic heart failure, ischaemic stroke, intracranial haemorrhage, ischaemic heart disease and CKD.

Concept — Primary prevention is **treatment of modifiable risk scaled to absolute risk**: the ASCVD score sets the threshold, LDL-C and blood pressure are the levers, and CAC scoring refines the intermediate-risk middle.

OBESITY AND WEIGHT REDUCTION

- Obesity is an important **modifiable cardiovascular risk factor**, defined by body mass index (BMI) categories of the National Institutes of Health and World Health Organization.

BMI CATEGORIES (KG/M²)

Category	BMI
Normal	18.5–24.9
Overweight	25–29.9
Obesity class I	30–34.9
Obesity class II	35–39.9
Obesity class III (severe)	≥ 40

- Prevalence of obesity and severe obesity in US men **has risen steadily** (2007–2008 to 2015–2016), reflecting unhealthy lifestyle and weight gain.
- Beyond BMI, **weight history** matters: in men, **gain of ≥ 5 kg after age 20 raises the risk of coronary heart disease and T2DM regardless of initial BMI**.
- **Comorbid cardiovascular risk factors:** hypertension, dyslipidaemia, raised triglycerides, impaired fasting glucose, obstructive sleep apnoea (OSA), tobacco use.
- **Noncardiac comorbidities** to screen: osteoarthritis, cholelithiasis, nonalcoholic fatty liver disease (NAFLD), depression, reduced quality of life.
- Consider weight-loss intervention at **BMI ≥ 25** , aiming to prevent, treat and reverse weight-related complications. **No evidence-based weight target exists**; recommendations are consensus-based.

WEIGHT MANAGEMENT GOAL BY BMI AND COMORBIDITY

Patient	Goal
BMI 25–29.9, no CVD or risk factors	Diet and exercise counselling to prevent further gain
BMI ≥ 30 , or 25–29.9 with CVD/risk factors, OSA or osteoarthritis	Achieve weight loss: behavioural modification, diet, exercise; more aggressive intervention at highest risk

- **Glucagon-like peptide-1 (GLP-1) receptor agonists** (e.g. semaglutide) provide a **drug-based option** alongside lifestyle modification and surgery; semaglutide is approved by the U.S. Food and Drug Administration (FDA) for chronic weight management as well as T2DM, and consumer demand has driven prescribing on a very large scale, including off label. Research on safety and efficacy in obesity continues.

Concept — **BMI classifies; comorbidity directs** — the intensity of weight intervention is set by attendant risk, and weight history can signal risk that current BMI hides.

GASTROENTEROLOGY IN THE AGEING MALE

- Gastrointestinal (GI) disorders are common and impair quality of life; in the ageing man consider **gastro-oesophageal reflux disease, constipation, diarrhoea, GI bleeding and colorectal cancer**.

Gastro-oesophageal Reflux Disease

- Gastro-oesophageal reflux disease (GERD) is **very common in US adults**; prevalence **rises with age**, and a substantial minority of sufferers have weekly symptoms.
- Reflux of gastric contents into the oesophagus **or beyond (oral cavity, lungs)**: pain, cough, wheeze, hoarseness, sore throat, oesophageal or chest fullness → distress, lost productivity and sleep.
- Mechanism: most likely **increased relaxation of the lower oesophageal sphincter (LES)**, worsening with age and comorbidity; **nonsteroidal anti-inflammatory drug (NSAID) use increases LES relaxation**.
- **Drugs damaging oesophageal mucosa**: NSAIDs, potassium, iron supplements, bisphosphonates.
- Typical presentation: **heartburn or acid regurgitation** — history can suffice for diagnosis. **Atypical symptoms prompt investigation**: chest pain, cough, wheeze, hoarseness, sore throat, intestinal symptoms, laryngitis, recurrent aspiration.

Evaluation: trial of **proton pump inhibitor (PPI) or histamine H2-receptor blocker** → symptoms persist → referral for **oesophageal pH studies or upper endoscopy**.

- Goals: **resolve symptoms and prevent complications**.
- **Lifestyle modification is the cornerstone**: weight loss, tobacco cessation, selective dietary restriction (coffee, highly spiced and acidic food), **elevate the head of the bed, no oral intake for 3 hours before lying down**, withdraw aggravating medications.
- Medical mainstays: PPIs and H2 blockers (both over the counter). **Long-term use → primary care or gastroenterology review** to identify **Barrett oesophagus** (pre-malignant) and mitigate oesophageal adenocarcinoma risk.

Chronic Diarrhoea and Constipation

- Both are exceptionally common in ageing men and **interface with male voiding dysfunction** → relevant to every urology provider.

CHRONIC DIARRHOEA VERSUS CHRONIC CONSTIPATION

	Chronic diarrhoea	Chronic constipation
Definition	Loose or liquid stools persisting ≥1 month	Infrequent or difficult passage of stool
Epidemiology	Common from the seventh decade; only a quarter of men seek help	Common in the general population; rises with age, especially in long-term care
Risk factors	Diabetes, prior GI surgery, polypharmacy, poor nutrition; acquired infection in long-term care	Age, long-term care residence; drugs including anticholinergics
Consequences	Declining mobility raises food insecurity → malnutrition and fluid imbalance	Poor perceived health, anxiety; faecal impaction, colonic pseudo-obstruction, urinary tract infection (UTI), urinary retention
Evaluation	History and examination usually find the cause; low threshold to refer	Detailed history, examination, possible abdominal imaging; screen for alarm features

- Patients seen for **UTI or urinary retention should always be evaluated for chronic constipation**.
- **Alarm features** (possible colorectal malignancy → **immediate gastroenterology referral for colonoscopy**): acute onset, weight loss, haematochezia, iron deficiency anaemia, family history of colorectal cancer or inflammatory bowel disease, positive faecal immunochemical test.

Constipation pathway: screen for alarm features (present → colonoscopy) → review drugs, including anticholinergics → seek secondary causes (inadequate dietary intake, emotional or physical abuse) → none found: therapeutic trial of **fibre or an osmotic laxative (polyethylene glycol)** → persisting → back to primary care or gastroenterology.

Occult Gastrointestinal Bleeding and Iron Deficiency Anaemia

- Occult GI bleeding is common and significant in ageing men; it can arise anywhere in the GI tract, and the **faecal occult blood test is often the primary indicator**.
- Blood loss → **iron deficiency anaemia (IDA)**, which carries a **more than thirtyfold increased risk of GI malignancy** compared with normal iron status.
- IDA rises with age — present in **more than a quarter of men over 85**.
- Anaemia → frailty, reduced physical performance, falls, cognitive decline, dementia, hospitalisation, death.

Colorectal Cancer Screening and Surveillance

- Colorectal cancer (CRC) is the **third most common cancer in men** and among the leading causes of male cancer death in the United States.
- Incidence rises with age and is **higher in males than females in every age cohort**.
- High incidence and mortality plus highly effective screening → **screening from age 45** (USPSTF and other bodies), **continued through age 75**; beyond 75, **shared decision making**.
- Recent decline in CRC mortality is attributed to improved screening over two decades.

Conclusions

- GI conditions are common in men, rise with age, impair quality of life and function, and are costly; they **often present with urologic disease**, so a working knowledge is needed by men's health providers.

Concept — Bowel and bladder **share risk factors, drugs and presentations**: constipation drives UTI and retention, anticholinergics cause constipation, and iron deficiency is a GI malignancy signal until proven otherwise.

DERMATOLOGY IN THE AGEING MALE

- About one-third of the US population sees a physician for a skin condition each year; burden rises with age, and those **65 or older have nearly double the skin problems** of younger patients.
- **Skin cancer is commonly detected at routine examination**; benign conditions also affect quality of life.

Skin Ageing

- Men show **accelerated skin ageing** and are perceived as older than women of similar age.
- **Photoageing** = age-related skin change worsened by ultraviolet (UV) light; **men are more susceptible** — more sunburn, half as likely to use sunscreen routinely.
- Signs: wrinkling, dyspigmentation, telangiectasia — collectively **dermatoheliosis**.
- UV **reduces collagen production and worsens collagen degradation**; ageing skin handles **reactive oxygen species** less well → inflammation; tobacco smoking adds further.
- **Prevention: daily sunscreen** (randomised controlled trials show it slows or prevents photoageing). **Treatment: retinoids** — epidermal thickening, increased collagen production, reduced UV-induced collagen degradation.
- **Impaired wound healing**: epidermal thinning plus elastin degradation → fragility, reduced tensile strength, slow healing.

- Epidermal thinning → **vitamin D deficiency** (epidermis is a primary site of synthesis).
- **Immunosenescence** of skin → reduced efficacy of skin-delivered vaccines and predisposition to skin cancer.

COMMON BENIGN GROWTHS

Acrochordon (skin tag)	Seborrhoeic keratosis
Pedunculated skin-coloured papule	Varied; classically round, demarcated, verruous surface
Together present in more than half of adults. Seborrhoeic keratosis needs no treatment but may be cosmetically bothersome, pruritic or inflamed. Both: liquid nitrogen cryotherapy or excision; histology if the diagnosis is uncertain.	

Skin Cancer

- **Commonest cancer in the United States:** over 2 million cases a year, **more than all other cancers combined**, and increasing (ageing population, better detection, more UV exposure).
- Risk factors: sun and tanning-bed exposure, recurrent sunburn, fair skin, northern European ancestry, long-term immunosuppression.

BASAL CELL CARCINOMA, SQUAMOUS CELL CARCINOMA AND MELANOMA COMPARED

	Basal cell carcinoma (BCC)	Squamous cell carcinoma (SCC)	Melanoma
Rank	Commonest skin cancer	Second commonest	Leading cause of skin cancer death
Sex	More common in men; rises with age	Men twice as likely; rises with age	Commoner in women when young, in men later; men fare worse (unexplained)
Appearance	Pearly papule that ulcerates and bleeds; mimics dermatitis	Pink papule or hyperkeratotic plaque , may ulcerate	ABCDE; de novo or from a naevus
Site	Sun-exposed skin	Sun-exposed head and neck	Chest and back; palms, soles, nail bed in darker skin
Metastasis	Exceptionally rare; locally destructive	Up to 5% → prompt treatment	Prognosis by stage
Diagnosis	Biopsy	Biopsy	Biopsy
Treatment	Excision including Mohs surgery; sometimes topical agents	Excision or topical agents	Stage-directed

- **ABCDE rule:** Asymmetry, Borders uneven, Colour variegated, Diameter >6 mm, Evolution over time.
- **Melanoma risk factors:** UV exposure, fair skin, **more than 50 naevi**, atypical naevi, a first-degree relative with melanoma, long-term immunosuppression.

Common Dermatoses

- **Xerosis** (dry skin): the **commonest dermatologic condition in ageing men**, affecting more than half of those 65 and over. Severe: red, cracked, adherent plate-like scale, pruritus; severe scaling = **asteatotic eczema (asteatotic dermatitis)**.
- Treatment: restore the **stratum corneum** and moisture — less bathing, emollients, avoid cold dry environments, humidify in winter; **topical steroids once eczema develops**.

DRUG ERUPTIONS: SIMPLE VERSUS SEVERE

Morbilliform (simple)	Severe
Appears 1–2 weeks after first exposure ; pruritic, not painful; low-grade fever common	Mucosal involvement , blistering, pain, erythema, oedema → consider Stevens-Johnson syndrome or toxic epidermal necrolysis
Commoner with age and medical comorbidity. Culprits: penicillin, cephalosporins, sulfonamide antibiotics, allopurinol, aromatic antiseizure drugs . Treatment: withdraw the offending agent , corticosteroids, antihistamines.	

- **Psoriasis**: equally prevalent in both sexes but **more severe in men**. Well-demarcated erythematous plaque with **silvery scale**; extensor forearms, shins, gluteal cleft.
- Treatment: topical steroids, calcineurin inhibitors, phototherapy → severe: **anti-tumour necrosis factor and newer biologics**.
- **Psoriasis is associated with CVD, including MI → query other CAD risk factors**; also with rheumatologic disease, including **psoriatic arthritis**.

Disorders of Skin Appendages: Nails and Hair

- Ageing nails: thicker, altered contour, texture and colour — probably from **poorer peripheral circulation and recurrent nail-bed trauma**.
- **Onychomycosis**: fungal nail infection, commoner in men and with age; risk factors peripheral vascular disease and diabetes. Nail-plate thickening, lifting from the bed, subungual debris.
- **Diagnosis before treatment**: curette debris → direct microscopy after **potassium hydroxide** → fungal hyphae.
- Treatment not mandatory; recommended for discomfort, infection or chronic immunosuppression. **First line: oral terbinafine 250 mg daily — 6 weeks fingernails, 12 weeks toenails**.
- **Alopecia** is a marker of generalised senescence, typically after 50. **Androgenetic alopecia** (male pattern baldness) affects up to half of men by 50; strong genetic component and established androgen role.
- First line: **5 α -reductase inhibitors** (reduce dihydrotestosterone) and **topical minoxidil** — **reduce or halt progression but do not restore lost hair**. **Hair transplantation** (follicle harvest and transfer) has advanced considerably.

Cosmetic Treatments and Procedures in the Ageing Male

- Men are increasingly high users — roughly a tenth of all **nonsurgical cosmetic procedures**, a marked rise.
- **Botulinum toxin**: the **most performed procedure in men**, for facial wrinkles; periorbital and forehead. Greater facial muscle mass → **lower doses may fail → higher doses → higher risk of brow ptosis**.
- **Soft tissue fillers** (second commonest): hyaluronic acid, calcium hydroxylapatite, poly-L-lactic acid, polymethyl methacrylate — soft-tissue augmentation and feature enhancement.
- **Topical retinoic acid is first-line for photoageing**: increases collagen, induces epidermal hyperplasia, decreases keratinocyte and melanocyte atypia; initial local irritation.

Conclusion

- Dermatologic conditions are common in men and cause morbidity and mortality: **benign conditions can be managed by men's health providers; severe conditions need immediate dermatology referral**.

Concept — **UV exposure and age generate most of the male skin burden**, and men photoage faster yet use less sunscreen — so prevention dominates, and triage divides benign disease managed in house from malignancy referred at once.

PSYCHIATRIC DISORDERS IN MALES

- Psychiatric disorders differ significantly in **prevalence, type and severity between the sexes**, partly through testosterone effects on brain structure and function.
- **Endogenous testosterone is strongly associated with mood disorders**, and exogenous testosterone may have a treatment role → men's health requires working psychiatric knowledge.

Testosterone and the Male Brain

- Lower male rates of depression and anxiety suggest a **protective effect of testosterone**.
- **Androgen receptors are dense in the hippocampus, amygdala and dorsolateral prefrontal cortex** — regions regulating mood, arousal, stress and memory.
- Prenatal and perinatal testosterone has **organisational and activational effects**; across life it influences **neuroplasticity and neurotransmitter regulation**.

SEX ASYMMETRY IN PSYCHIATRIC DISORDERS

Female-predominant (about twofold)	Male-predominant
Major depression	Substance use disorders (about twofold)
Anxiety	Autism (three- to fourfold)
Posttraumatic stress disorder	Attention-deficit/hyperactivity disorder (ADHD) (three- to fourfold)

- ADHD and autism may be associated with **higher prenatal testosterone**.
- **Opioids**: use has reached epidemic levels; **chronic opioid use causes testosterone deficiency** in about 5 million US men. **Endocrine Society**: screen men on chronic opioids for testosterone deficiency and evaluate for testosterone replacement therapy, which relieves symptoms in some.

Major Depressive Disorders

- Affect about 5% of the US population; **lower prevalence in men but markedly higher suicide**.
- **Suicide risk factors**: age, race, veteran status, psychosocial stress, **access to lethal means**, comorbid conditions → psychological screening and care in the multidisciplinary setting.
- Testosterone declines from middle age; **low testosterone in about a fifth of men in primary care**. It also falls with stress, medications, obesity, inactivity, diabetes and illness — **the same factors that raise depression**.
- **Symptom overlap** between testosterone deficiency and depression: low libido, irritability, anorexia, poor concentration, fatigue.
- Epidemiologic studies: **low testosterone associated with depressive symptoms and illness**; clinical trials of testosterone for depression **conflicting** (differing populations, doses, durations).
- Likeliest responders: **clearly testosterone-deficient, younger than 60, less severe depression**.
- **Testosterone is not recommended as routine treatment for depressive illness**, particularly major depression; responsive subgroups remain to be defined.

Conclusions

- Testosterone and its metabolites affect **neurogenesis, neurotransmitters, cerebral perfusion and metabolism**; testosterone falls with age and comorbidity; **deficiency is a risk factor for incident depression**, but therapy is not recommended because trial results conflict.

Concept — Association without proven treatment effect: low testosterone and depression share causes and symptoms, so the correlation is strong, but replacement is not a treatment for depression.

UROLOGIC DISORDERS OF FOCUS IN MEN'S HEALTH

- Topics on which urologists are the expert treating clinicians, covered in depth elsewhere: **testosterone deficiency, male infertility, ED and benign prostatic hyperplasia (BPH)** — the primary focus of care in the modern men's health centre.

Benign Prostatic Hyperplasia and Lower Urinary Tract Symptoms

- BPH = **nonmalignant enlargement of the prostate**; a predominant cause of **lower urinary tract symptoms (LUTS)** in ageing men; high prevalence → evaluation in both urology and primary care.
- Symptoms vary and may not be prostatic → **treatment focuses on symptom relief, not on a specific cause.**
- Management has shifted **from surgical to medical** → **primary care physicians now manage most patients.**
- **Refer to urology when medical therapy is insufficient, intolerable or not desired.** Transurethral resection or ablation remains a mainstay, with many newer minimally invasive technologies.

Erectile Dysfunction

- Affects **more than half of men aged 40–70.**
- Multifactorial → systematic **medical, surgical, psychosocial and exposure history** for diagnosis and a goal-oriented plan.
- **Every man with ED, whatever his age, is considered at risk of CVD**, even without other symptoms.
- Counsel all on **physical activity and diet**; some men and couples benefit from a **sexual health therapist.**

Treatment sequence: oral phosphodiesterase type 5 (PDE5) inhibitors (unless taking **nitrates** or physically unable to undertake sexual activity) → if inadequate or not tolerated, **local therapy**: intracavernosal injection of single or combined vasodilators, or intraurethral suppository; **vacuum erection device** as primary or adjunct therapy in selected men → **penile prosthesis**, which has the **highest satisfaction** and should be considered in all men with ED, after discussion of **irreversibility and implant loss from malfunction or infection.**

- **Non-FDA-approved therapies (platelet-rich plasma, low-intensity shock wave therapy) only within an experimental protocol.**

Male Infertility

- Definition: **failure to achieve pregnancy after 1 year of reasonable effort** (World Health Organization, 1991).
- Affects **15–20% of reproductive-age couples**; **male factors contribute in about half.**
- Evaluation: general and reproductive history **including the partner's age**; gonad-focused examination; **semen analysis — the cornerstone test.**
- Further testing as indicated: serum reproductive hormones, genetic testing, ultrasonography, cross-sectional imaging, specialised semen tests, testicular biopsy.
- **Some causes are treatable medically or surgically**; idiopathic cases may be neither identified nor treated → close partnership with **reproductive endocrinologists.**

Testosterone Deficiency and Hypogonadism

- Serum testosterone falls with age from **both testicular and hypothalamic-pituitary defects**; many attributed symptoms overlap with age-related comorbidity.
- Before testosterone replacement therapy (TRT): **exclude secondary causes** and perform **confirmatory testing.**

- **TRT criteria:** low serum testosterone confirmed on more than one occasion plus well-documented symptoms.
- **Randomised trial evidence:** improvements in obesity, metabolic syndrome, diabetes, muscle function and bone health — with **regular monitoring for adverse events**.
- Improvements in **mood, cognition and sexual function** rest on **observational data** and may be placebo effects.
- Multiple modalities and formulations, each with its own administration and monitoring plan.

Concept — *These four conditions are **why men come through the door**, and each is a window on systemic health — ED on cardiovascular risk, testosterone deficiency on metabolic disease.*

CONCLUSIONS

- **Convenience and access** have become central to health-care delivery.
- Men are **under-represented among seekers of preventive care**, yet carry **more comorbidity across most disease categories and die younger**.
- Men who avoid routine care **will seek care for problems "below the belt"** — sexual dysfunction, infertility, urinary control.
- The urologist keeps genitourinary expertise central but must hold a **fundamental understanding of disease in multiple organ systems** for initial evaluation and triage, backed by **referral networks and partners in other specialties**.

Concept — *The chapter turns on one asymmetry: **men avoid preventive care but not urologic care**, so the urologist becomes the entry point who must triage the whole man.*